



Advanced intrusion detection in the industrial Internet of Things using federated learning and LSTM models

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ARTICLE INFO

Keywords:

Anomaly detection
Cybersecurity in IIoT networks
CNN-Bi-LSTM
Federated learning (FL)
Industrial Internet of Things (IIoT)
Intrusion detection system (IDS)
Privacy-preserving security

ABSTRACT

The growing prevalence of Industrial Internet of Things (IIoT) networks has intensified the demand for robust and scalable intrusion detection mechanisms. However, data privacy and protection with high detection accuracy remains a challenge. Centralized frameworks typically decentralize privacy and are not scalable. This study aims to propose a hybrid Convolutional Neural Networks - Bidirectional Long Short-Term Memory (CNN-Bi-LSTM)-based intrusion detection framework integrated with Federated Learning (FL) to address these challenges in IIoT environments. The framework combines the spatial feature extraction capabilities of Convolutional Neural Networks (CNNs) with the temporal dependency analysis of Bidirectional Long Short-Term Memory (Bi-LSTM) networks. To validate the framework, experiments were conducted using three benchmark IIoT datasets: X-IIoTID, WUSTL-IIoT, and Edge-IIoTset. Data preprocessing included normalization, feature encoding, and class balancing. FL was employed to preserve data privacy, with local models trained on edge devices and aggregated at a central server. The model's performance was evaluated against centralized learning approaches, considering accuracy, precision, recall, F1-score, and Area Under the Curve (AUC) metrics. The proposed framework demonstrated superior intrusion detection capabilities with an accuracy of 97.8 % on X-IIoTID and consistent AUC scores above 0.95 across all datasets. FL significantly reduced communication overhead while ensuring data privacy. The hybrid architecture detected diverse attack types, including Distributed Denial of Service (DDoS), Injection, Backdoor, and Man-in-the-Middle (MitM) attacks. The hybrid CNN-Bi-LSTM model and the FL offer a way of securing the IIoT on a scalable and private level. The results show the possibility of combining progressive AI approaches with decentralized structures to strengthen IIoT security and effectiveness.

1. Introduction

The IIoT improvised the industry through process automation and integrating smart devices, sensors, or networks. In contrast with other conventional methods, IIoT interrelates tangible devices to the internet; thus, the IIoT environment is a taxing network that can provide smooth media for interaction and the transfer of information [1]. This technological advancement has many advantages, but, at the same time, it creates weaknesses that make these networks ideal for cyber-attacks. Security risks that include DDoS attacks, data loss, and malware threats expose IIoT foundations to considerable perils that halt operations and threaten private data [2]. Such threats have been fought using

traditional IDS with centralized data processing and analysis [3]. However, centralized methods have several drawbacks as the size and complexity of IIoT environments evolve. First, they have a problem managing vast quantities of the data collected at the edge devices; this leads to latency and inefficiency. Secondly, the confinement of data for analytical purposes is a troublesome idea because it centralizes the data that can be, and in many industrial uses of AI, highly sensitive information [4]. These limitations indicate that more adaptive, scalable, and privacy-preserving solutions are the need of the hour. Federated Learning (FL) has now become prominent as a solution to these challenges [5]. FL refers to allowing multiple devices to train the machine learning model collectively but without sending their original data.

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<https://doi.org/10.1016/j.adhoc.2025.103991>

Received 11 February 2025; Received in revised form 26 June 2025; Accepted 31 July 2025

Available online 5 August 2025

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However, only model updates are transferred, making raw data transmission local and avoiding information leakage. This is beneficial in improving data privacy and decreases the overall communication overhead in IIoT networks [6].

Intrusion detection performance is, therefore, enhanced, and more FL frameworks are increasingly deploying advanced deep learning techniques. A combination of CNN and Bi-LSTM network models has proven highly effective. CNNs work well at decoding spatial features from the network traffic. At the same time, Bi-LSTM models perform well in identifying temporal relations, which is important in the sequential characteristics of attacks. These architectures offer a strong paradigm for recognizing secure and novel attacks in IIoT networks [4]. As the use of IIoT technologies intensifies, some chances go along. The IIoT systems are improving connectivity and automation in industries, but they are also exposed to sophisticated cyber threats. These threats include DDoS attacks, data breaches, and malware, which are security and operation reliability threats. It is argued that the more centralized architectures based on which Traditional Intrusion Detection Systems (IDS) are becoming less well-suited to dealing with these new issues [7]. Inherent disadvantages of centralized IDS architecture include latency, scalability problems, and single-point failure. Massive collection of large amounts of sensitive information to a server for processing amplifies these issues, requiring greater bandwidth and exposing the information to privacy violations. Compared to traditional systems where the data is centralized and processed within certain related facilities, these shortcomings are amplified within IIoT networks that use generated and processed data in distributed environments [8]. Due to these problems, FL has devised an effective way to solve them. FL also improves privacy and lowers communication costs since detailed data does not need to be shared to train the model while using a decentralized approach. However, the model architecture dramatically influences the performance of FL-based IDS frameworks. FL has applied basic machine learning models, but due to their inherent structure, they cannot effectively recognize various step-by-step attacks [9].

CNNs and BiLSTMs architectures showed high performance when used to classify texts of this nature. CNNs are accurately applicable to decomposing spatial features from the network traffic, and the Bi-LSTM networks are good at extracting temporal patterns and, therefore, are more appropriate for recognizing intricate patterns of attacks. While these hybrid models have the potential of the FL, their integration into FL frameworks for IIoT intrusion detection has drawn little research attention so far [10]. This research presents a new FL-based IDS architecture with a CNN-Bi-LSTM structure to fill these gaps. In this study, we aim to implement a framework that handles many IIoT datasets and is accurate and privacy-preserving in our approach toward identifying numerous new cybersecurity threats within IIoT networks.

The main objective of this study is to propose a novel high-performance IDS for IIoT networks that considers the three critical factors of scale, accuracy, and privacy. In this work, we suggest using FL to improve the effective detection of known and novel threats in the IIoT context using a hybrid CNN-Bi-LSTM model. The research aims to solve the problems of high costs, poor detection, and centralized IDS by using the FL model to provide data privacy and achieve high detection performance.

- To design a scalable and privacy-preserving IDS architecture tailored for IIoT networks, enabling decentralized model training without transferring raw data.
- To combine the spatial feature extraction capabilities of CNNs with the temporal pattern recognition strengths of Bi-LSTM networks to improve the detection of sophisticated cyberattacks.
- To test the proposed framework using diverse IIoT datasets to assess its effectiveness in detecting various attack types, including Distributed Denial of Service (DDoS), backdoor intrusions, and malware.

- To analyze the advantages and limitations of centralized vs. federated IDS frameworks regarding accuracy, latency, and resource efficiency.

This work presents the following novel contributions to the concepts of cybersecurity and IIoT. It proposes a new FL-based solution suitable to counteract the security threats arising in IIoT systems. Using both CNN and Bi-LSTM within the FL scheme, the proposed solution substantially improves the ID process with high accuracy and adjusts for new intrusions. In terms of various attack situations and the network topology, the performance of this study utilizes benchmark datasets, including X-IIoTID [2], WUSTL-IIoT [11], and Edge-IIoTset [4]. Furthermore, the solution focuses on data privacy in its implementation, as the models are trained only on edge devices. Furthermore, comparing centralized and federated IDS can provide significant information that would benefit researchers and practitioners who desire to strengthen the protection and availability of IIoT networks. To this end, this research seeks to meet the following objectives: enhance the existing state of IDS for IIoT and develop a more effective solution against emerging threats.

The rest of the paper is organized as follows: The related work section provides a detailed literature review, covering existing intrusion detection methods and the role of FL in IIoT security. The proposed methodology section describes the datasets used and the preprocessing techniques applied. This section also presents the proposed methodology, including the CNN-Bi-LSTM architecture and FL framework. The experimental setup details the tools, configurations, and hyperparameters. The results and analysis section discusses the findings, highlighting the model's performance across datasets and attack types. The conclusion and future work section concludes the paper with a summary of findings, contributions

2. Related work

2.1. Intrusion detection systems for IIoT

IDS defends IIoT networks against cyber threats. These systems watch out for network traffic patterns and detect risks that illegitimate users may cause. In the IIoT setting, IDS has been seen as more imperative owing to the disaggregate devices and the intricate traffic flow [12]. The conventional IDS approach mainly consists of two detection methodologies: signature-based and anomaly-based [13]. The most common type of IDS is the signature-based IDS, which uses predefined rules or patterns to check out for known attacks. Compared to the new and emerging types of risks, they perform well if an organization has documented risk profiles to compare. While anomaly-based systems detect odd behavior from the norm, the system's functionality is focused on the program's normal behavior. These systems are more effective when used in an environment where the threats are unknown and can detect the threat more quickly; however, they have more false positives than true positives [14]. For the IIoT networks, the centralized IDS architectures have been widely implemented. They collect data and information from different devices and store it on a single server for computation. Although such systems appear centralized, they cause issues related to scalability and latency when the number of connected devices increases. Also, through centralized data processing, privacy issues emerge, which is very important when data security is necessary [15]. Machine learning (ML) and deep learning (DL) enhanced a new generation of IDS. Classification has been performed on network traffic using ML algorithms like Random Forest (RF) and Support Vector Machines (SVM). However, they usually fail when dealing with large and complex IIoT data. HDL models are further improved with better performance of DL models such as CNN and LSTM networks. CNNs can extract features from raw data, while the LSTMs learn temporal properties of the network traffic [16]. Nevertheless, there are still difficulties that have to be solved. Centralized approaches to implementing basic IDS using ML and DL sometimes need data collection at one central

point, implying privacy and latency inconvenience. These challenges have led to exploring decentralized approaches such as FL, which allows modeling and training without transmitting raw data. While analyzing the experiences of deploying FL in IIoT networks, FL has demonstrated potential in handling privacy concerns while achieving high detection identification in [17,18].

The changing nature of IDS enhances the need to adopt novel methodologies, including hybrid models and FL frameworks, to the IIoT. They can improve the IIoT network security and reliability since they resolve the deficiencies of the centralized, conventional solutions.

2.2. Centralized vs. federated learning in IIoT networks

As for the security issue, the IIoT network's machine learning is suitable for IDS. These models' two principal training categories include centralized and FL, with their special characteristics. In the centralized learning paradigm, the raw data is collected across different edge devices to perform training of machine learning models. This simple technique enables the server to employ various data sets to enhance the detection accuracy [19]. However, the centralized learning approach has some inherent drawbacks that prevent the application of this concept to IIoT environments. Firstly, because the amount of data being generated within IIoT systems is vast, there is a considerable demand for bandwidth. Relaying this data to a central server results in network congestion and latency problems, particularly in real-time applications [20]. Secondly, keeping information in one place is rather dangerous regarding privacy. It can, therefore, be said that a breach at the central server threatens confidential industrial operations since data and system integrity may be affected [21]. These challenges make it difficult to scale and apply centralized learning for contemporary IIoT systems. FL provides a distributed solution where raw data is stored at the edge devices, and only the models are synchronized with a server. This method is more private and requires less data to be transferred across networks than other methods [22,23]. FL is most applicable in the IIoT context since the devices involved will be widely distributed, and the privacy of the data transmitted will be an issue. FL reduces the probability of data leaks since models are trained locally, and updates are combined. It also cuts across the bandwidth facility in a way that diminishes the usage of this basic need, especially for scarce networks [24]. All the same, FL has its own set of challenges that are worth discussing. The frequent changes in the working dataset on different gadgets must be in harmony, and the distribution of data can also influence a model's structure and convergence speed. Additionally, the computational demands at the edges might burdensome low-resource IIoT systems [25]. Centralized learning is still beneficial for cases where data and information-sharing concerns are not so critical. However, several drawbacks disqualify it from large-scale, privacy-concerned IIoT applications. On the other hand, FL is a good solution to achieve IIoT security with scaling and privacy concerns being mitigated. This paper selects FL as a new IDS model framework and combines CNN-Bi-LSTM to identify advanced threats targeting IIoT systems. Therefore, the solution proposed here combines FL's decentralization with the feature extractors and sequential analysis of CNN and Bi-LSTM, respectively, with an expected high detection accuracy while maintaining clients' data privacy and system efficiency.

2.3. Role of CNN-Bi-LSTM in anomaly detection

In anomaly detection for IIoT networks, there is considerable interest in predicting and detecting unusual behavior using CNN and Bi-LSTM deep learning models. These models combine CNN's effectiveness for spatial feature recognition and Bi-LSTMs for temporal sequence learning, offering a strong basis for studying intricate attack scenarios. CNNs are used to learn from raw traffic data, such as network traffic or packet-level statistics, and are designed to learn spatial patterns and correlations. This makes them suitable for identifying localized features, such as flows of different packets and/or features in the IIoT network

that are different from the expected baseline. Using convolutional layers, CNNs can detect features that are very hard for humans to detect and can save much time, especially in the high number of feature spaces [26]. In anomaly detection tasks, CNN extracts the valuable features that act as a basis for the subsequent analysis by the sequential models. Significantly, Bi-LSTM networks are beneficial for analyzing sequential data, which makes them capable of discovering temporal dependencies of IIoT traffic. They differ from the traditional LSTMs that take data in one direction; Bi-LSTMs analyze data in two directions, so they correlate past and future states within the given network. This bidirectional capability helps identify temporal patterns inherent to threats such as DDoS or steady backdoor intrusion [27]. The CNN Bi-LSTM architecture is a fusion of the CNNs for feature extraction and the Bi-LSTM for analyzing sequential data. This architecture works in two stages: In the first CNNs, spatial features are extracted from the raw traffic data, while Bi-LSTMs process the data to identify temporal correlations. The presented layered model allows for detecting space and time anomalies crucial for analyzing complex cyber threats in IIoT networks [28]. Also, the combined models are more appropriate for solving the problem of an unbalanced dataset, which is characteristic when implementing IIoT systems where the traffic labeled as malicious accounts for a minuscule proportion of the entire data. Thus, using both spatial and temporal dimensions, CNN-Bi-LSTM makes acceptable detection results with better suppression of false positives to facilitate large-scale and dynamic IIoT networks [29]. Recent studies have unveiled various side-channel attacks exploiting electromagnetic emissions and other physical phenomena in IoT devices. For instance, Ni et al. demonstrated that electromagnetic emissions during wireless charging can be exploited to infer user interactions on smartphones [30]. Similarly, Zhang et al. showed that electromagnetic side channels could be leveraged to recover fingerprints from in-display sensors, raising concerns about biometric data security [31]. In another study, Ni et al. explored how side channels in wireless charging power banks can lead to privacy inference through few-shot learning, emphasizing the vulnerabilities in modern IoT infrastructures [32]. Additionally, Zhao et al. investigated the security implications of multi-port chargers, demonstrating how they can be exploited for privacy leakage and voice injection attacks [33]. Furthermore, Ni et al. revealed that radio-frequency energy harvesting could be a vector for eavesdropping on mobile app activities, highlighting new risks in IoT environments [34]. Beyond anomaly detection, deep learning-based authentication mechanisms have been widely explored to strengthen IoT security. Cao et al. introduced a passive vibration response-based authentication system, leveraging hand biometrics for non-intrusive user verification [35].

In addition to CNN-Bi-LSTM models, FL strengthens it when integrated. FL allows joint training over multiple devices while raw data remains on each device, protecting users' privacy and minimizing communication overhead. FL's framework generalization compatibility with other anomalous heterogeneities correlates with the hybrid model's effectiveness in this context without compromising scalability or privacy [36].

3. Proposed methodology

The proposed methodology introduces a hybrid CNN-Bi-LSTM model integrated with an FL framework for intrusion detection in IIoT networks. This approach leverages CNN's spatial feature extraction capabilities and the temporal dependency modeling of Bi-LSTM to detect complex attack patterns. The FL framework ensures data privacy by enabling collaborative training across distributed devices without sharing raw data. Fig. 1 depict the proposed methodology.

3.1. Overview of selected datasets

This study employs three widely recognized datasets to evaluate the proposed intrusion detection framework: X-IIoTID, WUSTL-IIoT, and

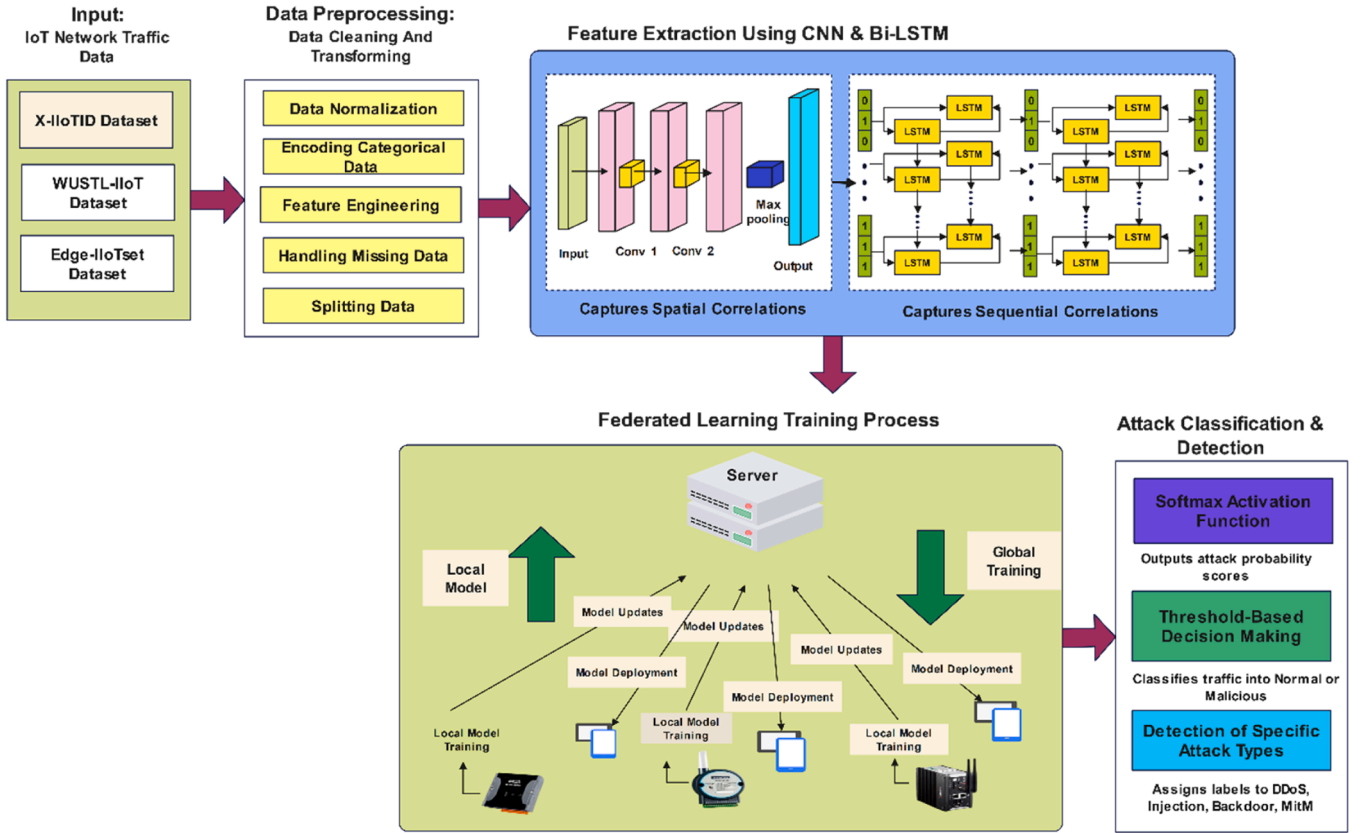


Fig. 1. Proposed methodology.

Edge-IloT set. Comprising different and wholesome information, these datasets capture the actual IloT complexities with varying scenarios of attack and traffic norms captured.

3.1.1. X-IloTID dataset

The X-IloTID comprises data required to detect intrusion on the IloT systems. It is not specific to hardware, making achieving good generalization in any industrial environment possible. This contains millions of records; it covers multiple attacks, including DDoS, Man-in-the-Middle (MitM), unauthorized backdoor entrance, and port scanning attacks. The dataset also includes operational traffic, which involves device-related activities. Some of the variables include packet level parameters, connection properties, and payload information, and as such, it is an excellent source of data for developing machine learning algorithms.

3.1.2. WUSTL-IloT dataset

The second important dataset employed in the current study is the WUSTL-IloT dataset. It delivers network vulnerability information and includes live and emulated threat information. The data set has traffic records that contain attacks: malware spread, ransomware attacks, data theft, and DDoS. It also records the general traffic activity that may result from regular IloT sensors to control interactions. This dataset includes native flow-level statistics, timestamps, and info regarding the source and destination of specific flows, which are critical for determining machine learning-based counter-measures effectiveness.

3.1.3. Edge-IloTset dataset

Edge-IloTset consists of the Edge-IloT dataset with modifications to mimic a real IloT environment, emphasizing cybersecurity at the network edge. Concerning the type of attacks, this dataset has HTTP and ICMP-based flood attacks, backdoor attacks, and injection attacks. Traffic in this dataset is characterized by regular traffic, which refers to

communication between industrial controllers and edge devices. The data set consists of packet payload records, traffic flows, and attack samples and is the most comprehensive of all potential threats for IloT networks.

The summary of these datasets is presented in Table 1. Each dataset is preprocessed to ensure consistent and reliable performance during evaluation.

3.2. Data preprocessing and feature engineering

Data preprocessing and feature engineering are necessary steps across both sets of datasets as a key foundation for preparing data for model training and assessment. Such measures help enrich feature representation and manage data-related issues like noise, imbalance, or high dimensionality. The observational records within the data sets are preprocessed to remove duplicates, missing values, or noise records that are not relevant. Data has been preprocessed for features that contain missing values: mean or median imputation for continuous data and mode imputation for categorical data. For example, given a feature x with missing values, imputation is performed as:

$$x_{filled} = \begin{cases} \mu_x & \text{if numerical} \\ mode(x) & \text{if categorical} \end{cases} \quad (1)$$

where μ_x represents the mean of x , and $mode(x)$ is the most frequent value. To ensure consistent scaling across features, normalization is applied using min-max scaling. Each feature x_i is transformed into the range $[0, 1]$ as follows:

$$x' = \frac{x_i - \min(x)}{\max(x) - \min(x)} \quad (2)$$

The transformation makes it possible to remove the effect of features with large values in datasets compared to features with small ranges

Table 1
Summary of datasets.

Dataset	Records	Attack Categories	Normal Traffic	Key Features	Source
X-IIoTID	~6000,000	DDoS, MitM, Backdoor, Port Scanning	Device-specific communications	Packet-level attributes, payload metadata	[2]
WUSTL-IIoT	~2500,000	Malware, Ransomware, Data Exfiltration, DDoS	Interactions between sensors/controllers	Flow-based stats, timestamps, identifiers	[11]
Edge-IIoTset	~8000,000	DDoS-HTTP, DDoS-ICMP, Backdoor, Injection Attacks	Industrial controllers and edge devices	Packet payload, traffic patterns	[4]

when training a model. The attack labels and other categorical variables are then transformed into numbers using one-hot or label encoding. For example, an attack type $T \in \{Normal, DDoS, Injection\}$ is encoded as:

$$One - Hot(T) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (3)$$

This representation also helps to make compatibility with machine learning models. There are reference datasets from IIoT systems, and they often have a problem with class imbalance, having fewer examples of attacks than normal traffic. To this end, the Synthetic Minority Oversampling Technique (SMOTE) is used to oversample the minority class. SMOTE generates synthetic samples as follows:

$$x_{new} = x_i + \delta \cdot (x_j - x_i) \quad (4)$$

where x_i and x_j are samples from the minority class and δ is a random number in $[0, 1]$. Feature selection uses statistical methods such as correlation analysis or mutual information. For example, Pearson correlation r between two features x and y is computed as:

$$r = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{\sqrt{\sum (x_i - \mu_x)^2 \sum (y_i - \mu_y)^2}} \quad (5)$$

Features with low correlation to the target variable or high multicollinearity are excluded to reduce dimensionality.

3.3. Attack classifications in the datasets

This study represents all these attack types with diverse datasets that capture different behaviors and patterns. These classifications are of paramount importance in developing a sound intrusion detection framework. It is important to note that significant packet rates and small time between arrivals in networks of IIoT confirm high traffic. Injection attacks take advantage of input injection, including SQL or command injections, and are usually identified by other features of a payload containing unfamiliar characters. The backdoor attack means that the violator's unauthorized access is supported by permanent connections or atypical ports distinguishable by high connection rates during a definite time range. MitM attacks deceive the gateway, recognizable by automatic communication sequence breaks or cryptographic checks. Knowledge of these categories helps to detect and protect effectively.

Table 2 shows the percentage of regular and attack classes within the datasets. The variability of the population ensures that different circumstances challenge the model.

Table 2
Summary of the regular and attack classes distribution.

Dataset	Regular Traffic (%)	DDoS (%)	Injection (%)	Backdoor (%)	MitM (%)
X-IIoTID	60	20	10	5	5
WUSTL-IIoT	50	25	15	5	5
Edge-IIoTset	55	30	10	3	2

The datasets are prepared to train the hybrid CNN-Bi-LSTM model effectively by employing robust preprocessing techniques and leveraging feature engineering.

3.4. Federated learning framework

The advocated methodology is based on FL and a CNN-Bi-LSTM model for detecting intrusion in IIoT networks. In this framework, data is kept on local edge devices, which makes the data private. Each device runs a regional model on its data and exchanges only the model updates with a central aggregator. These updates are accumulated at the global server to develop a unified model propagated back to the devices for rebuffing. This iterative process allows for excellent learning of parameters that can be used in shared tasks without sharing the raw data, causing privacy issues, or being unscalable.

CNN component took spatial features from the IIoT traffic, while the Bi-LSTM took temporal aspects of the traffic to make a strong hybrid model to detect different attacks. This arrangement helps enhance detection in various and distributed IIoT systems.

3.5. CNN-Bi-LSTM model architecture

CNN-Bi-LSTM is a model that combines CNN's pervasive feature extraction proficiency in the spatial domain with the sequential analysis efficiency of Bi-LSTM networks. This time- and space-correlated pattern analysis is a feature of this hybrid architecture explicitly intended to identify intricate attack schemes in IIoT networks based on traffic data. These layers are the descriptions of the model architecture. The CNN-Bi-LSTM model architecture embraces several interconnected layers that serve different intrusion detection purposes. The Input Layer takes initial network traffic characteristics, for instance, packet features and flow characteristics that presumably are independent records, and formats them as tensors for deep learning. The CNN Layer performs spatial features from this input data and detects region-wise patterns to predict anomalous conditions. To increase running speed, the Pooling Layer also decreases the size of features the system needs to work with. Subsequently, the Bi-LSTM Layer takes these spatial features to capture temporal dependencies in the network traffic, where the bidirectional characteristics of the Bi-LSTM Layer depend on both past and future states. The Fully Connected Layer combines input from the Bi-LSTM to create a feature representation to classify the videos. Also, the final layer, the Output Layer, will use a SoftMax activation function to classify the network traffic as usual or an attack to end the detection process. These diverse layers of abstraction are beneficial for accurate and effective anomaly detection in the IIoT setting.

Algorithm 1 outlines the process of training a CNN-Bi-LSTM model on IIoT datasets for accurate anomaly detection.

3.6. Centralized and federated training approaches

The IDS in IIoT considerably relies on the type of training technique employed. This study analyzes the strengths and weaknesses of centralized and FL strategies.

Algorithm 1

CNN-Bi-LSTM for intrusion detection.

```

1. Input: Dataset  $D = (X, Y)$ , Number of epochs  $E$ , Batch size  $B$ , Learning rate  $\alpha$ 
2. Output: Trained CNN-Bi-LSTM model  $M$ 
3. Initialize the CNN-Bi-LSTM model parameters  $W_{conv}, W_{lstm}, W_{fc}, b_{conv}, b_{lstm}, b_{fc}$ 
4. Preprocess dataset  $D$ : Normalize features  $X$ , encode labels  $Y$ , and split into training  $D_{train}$ , validation  $D_{val}$ , and testing  $D_{test}$ .
5. For epoch  $e = 1$  to  $E$ :
6.   Shuffle  $D_{train}$  into mini-batches of size  $B$ .
7.   For each mini-batch  $(X_b, Y_b)$  in  $D_{train}$ :
8.     Pass input  $X_b$  through the convolutional layer:
9.      $F = \text{ReLU}(\text{Conv}(X_b; W_{conv}) + b_{conv})$ 
10.    Apply max-pooling to reduce feature dimensionality:
11.     $P = \text{MaxPooling}(F)$ 
12.    Pass  $P$  through the Bi-LSTM layer to capture temporal dependencies:
13.     $H_t = \text{Bi-LSTM}(P_t; W_{lstm}, b_{lstm})$ 
14.    Pass  $H_t$  through the fully connected layer:
15.     $Z = \text{ReLU}(W_{fc} \cdot H + b_{fc})$ 
16.    Generate predictions using the SoftMax layer:
17.     $\hat{Y} = \text{Softmax}(Z)$ 
18.    Compute loss using cross-entropy:
19.     $L = -\frac{1}{B} \sum_{i=1}^B Y_i \log(\hat{Y}_i)$ 
20.    Update weights  $W_{conv}, W_{lstm}, W_{fc}$  and biases using backpropagation and the Adam optimizer.
21.   End For
22. Evaluate  $M$  on  $D_{val}$  and adjust hyperparameters if necessary.
23. End for
24. Test  $M$  on  $D_{test}$  and compute accuracy, precision, recall, and F1-score metrics.
25. Return: Trained CNN-Bi-LSTM model  $M$ .

```

3.6.1. Centralized training

Another type of solution is centralized training, meaning all the data collected in distributed devices is sent to the central training server. The data is then fed to the central server to develop a global model that will help detect intrusion patterns on the IIoT. In centralized training, the common process follows a systematic approach to develop and assess the intrusion detection deep learning model. The first stage is Data Collection, where data collected from all IIoT devices is transported to a unified database to form a records database. Next, in the Model Training phase, the expanded dataset again trains a deep learning model like CNN-Bi-LSTM that delivers an anomaly's accurate spatial and temporal details. Last, the evaluation phase checks the performance of the trained model in different testing datasets, and it uses accuracy, precision, recall, and F1-score to ensure the model's efficiency in detecting intrusions. Ideally, centralized training yields high detection accuracy because the system can access all the requisite data. However, it suffers from significant limitations, such as privacy concerns, high communication overhead, and vulnerability to single points of failure. These disadvantages affect its suitability for IIoT systems because privacy and scalability requirements are mandatory in such systems.

3.6.2. Federated training

Federated training works to overcome the shortcomings of centralized systems by providing a setup in which the model training is performed locally on devices with only the parameters for the model update, e.g., gradients, being exchanged with a central server. The server aggregates these updates to form a global model, which is then distributed back to devices for further refinement. [Algorithm 2](#) outlines a privacy-preserving process where local models on distributed devices are trained on private datasets, and their updates are aggregated at a central server to refine a global model.

3.7. Evaluation metrics

Various evaluation metrics are employed to assess the performance of the proposed CNN-Bi-LSTM intrusion detection framework. These metrics provide insights into the model's accuracy, precision, ability to recall attacks, and effectiveness in detecting anomalies in IIoT networks.

Algorithm 2

Federated training approach.

```

1. Input: Distributed datasets  $\{D_k = (X_k, Y_k)\}_{k=1}^K$  on  $K$  devices, Number of global rounds  $G$ , Local epochs  $E$ , Batch size  $B$ , Learning rate  $\alpha$ .
2. Output: Trained global model  $M_{federated}$ .
3. Initialize the global model  $M_{global}$  parameters  $W_{global}, b_{global}$ .
4. For the global round  $g = 1$  to  $G$ :
5.   Distribute  $M_{global}$  to all  $K$  devices.
6.   For each device  $k = 1$  to  $K$  (in parallel):
7.     Train local model  $M_k$  for  $E$  epochs on  $D_k$ :
8.     Shuffle  $D_k$  into mini-batches of size  $B$ .
9.     For each mini-batch  $(X_{b,k}, Y_{b,k})$  in  $D_k$ :
10.      Forward pass through the local model to compute predictions  $\hat{Y}_{b,k}$ .
11.      Compute the local loss:
12.       $L_k = -\frac{1}{B} \sum_{i=1}^B Y_{i,k} \log(\hat{Y}_{i,k})$ 
13.      Backpropagate the loss and update local parameters  $W_k, b_k$ .
14.     End For
15.     Send local updates  $\Delta W_k, \Delta b_k$  to the central server.
16.   End For
17.   Aggregate local updates at the server:
18.    $W_{global} = \frac{1}{K} \sum_{k=1}^K W_k, b_{global} = \frac{1}{K} \sum_{k=1}^K b_k$ 
19.   Update  $M_{global}$  with aggregated parameters  $W_{global}, b_{global}$ .
20.   Evaluate  $M_{global}$  on a global validation set (optional).
21. End For
22. Test  $M_{federated}$  on a global test set and compute metrics.
23. Return: Trained global model  $M_{federated}$ .

```

Accuracy measures the proportion of correctly classified instances, both normal and attack traffic, to the total number of cases. It is defined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (6)$$

where:

- **TP:** True Positives (correctly identified attacks)
- **TN:** True Negatives (correctly identified normal traffic)
- **FP:** False Positives (normal traffic misclassified as attacks)
- **FN:** False Negatives (attacks misclassified as normal traffic)

Precision evaluates the proportion of correctly identified attacks out of all instances predicted as attacks. It emphasizes the relevance of predictions.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (7)$$

Recall measures the model's ability to identify attacks among all attack instances. A higher recall indicates fewer false negatives.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (8)$$

The F1-score is the harmonic mean of precision and recall, providing a balanced measure that accounts for false positives and negatives.

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (9)$$

TPR, also known as sensitivity, represents the ratio of correctly identified attack instances to the total attack instances.

$$\text{TPR} = \frac{TP}{TP + FN} \quad (10)$$

FPR quantifies the proportion of regular traffic instances incorrectly classified as attacks.

$$\text{FPR} = \frac{FP}{FP + TN} \quad (11)$$

RMSE evaluates the difference between predicted and actual values, offering insights into the model's error magnitude. It is calculated as:

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2} \quad (12)$$

Where $\hat{y}_i$ and y_i are the predicted and actual labels, respectively, and N is the total number of instances. The effectiveness of the discussed training paradigms is assessed by comparing the above metrics on test datasets of both centralized and federated training methodologies. The work examines the costs and benefits of accuracy, precision, recall, and processing time to conclude how IIoT anomaly detection training should be carried out.

The federated training improves privacy because raw data is stored locally on devices; this minimizes data privacy leakage. It also scales better in large IIoT networks because it saves more frequent communication overhead. However, federated training may suffer from such limitations as device heterogeneity, the data imbalance challenge, and the synchronization challenge.

3. 8 Experimental setup

The experiments were conducted on a high-performance system equipped with an NVIDIA GPU and 32 GB of RAM, utilizing TensorFlow as the core deep learning framework. Python served as the primary programming environment for data preprocessing, model development, and performance evaluation. Libraries such as Scikit-learn and Matplotlib were employed for feature selection, data visualization, and analytical interpretation. Three benchmark IIoT datasets, X-IIoTID, WUSTL-IIoT, and Edge-IIoTset, were selected due to their rich diversity in attack types and their relevance to real-world industrial environments. Prior to training, the datasets were preprocessed through normalization and class balancing techniques. A standard 80:10:10 split was used for training, validation, and testing, ensuring consistency in comparative evaluation. Model training utilized the Adam optimizer with a learning rate of 0.001 and a batch size of 64. Training was carried out over 50 epochs, with early stopping based on validation loss to mitigate overfitting. Baseline hyperparameter tuning determined the best architecture configuration, resulting in 32 CNN filters, 128 Bi-LSTM units, and a dropout rate of 0.3, providing a balance between performance and computational efficiency. In the FL setup, training was distributed across multiple clients, with 50 communication rounds conducted between local models and the central aggregator. While FL avoids raw data transmission and enhances privacy, we observed that the cumulative communication overhead, especially given model sizes of 70–80 MB—can approach or exceed the original dataset size (~100 MB) after many rounds.

4. Results and analysis

4.1. Performance on individual datasets

The proposed CNN-Bi-LSTM model was tested on the X-IIoTID, WUSTL-IIoT, and Edge-IIoTset databases. The first and second datasets included different traffic and attack patterns, which allowed for a thorough evaluation of the models to test their ability to generalize in different IIoT settings.

We got the maximum correct classification accuracy in every set using this model. The accuracy results are summarized and presented graphically in Fig. 2.

Fig. 3 depicts precision, recall, and F1-scores for X-IIoTID, WUSTL-IIoT, and Edge-IIoTset datasets, demonstrating they are almost equally affecting and balancing all scores of the model.

Confusion matrices were generated for each dataset, visualized as heatmaps. These matrices show True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). Fig. 4 heatmaps of confusion matrices for X-IIoTID, WUSTL-IIoT, and Edge-IIoTset datasets highlight classification performance.

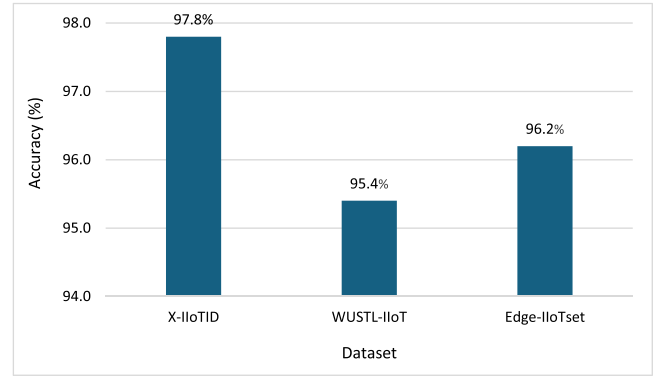


Fig. 2. Accuracy comparison across datasets.

Fig. 5 depict the TPR and FPR across X-IIoTID, WUSTL-IIoT, and Edge-IIoTset datasets. It highlights the model's ability to detect attacks (high TPR) while minimizing false alarms (low FPR).

Training, validation loss, and accuracy over epochs were generated to visualize model convergence for each dataset. Fig. 6 depict training, validation loss, and accuracy curves to demonstrate the model's convergence and stability.

ROC curves were generated for each dataset, and the Area Under the Curve (AUC) scores are provided in the table below. Fig. 7 shows the model's ability to distinguish between regular and attack traffic.

Feature importance was analyzed to identify the most influential features for classification. Fig. 8 plots show the top 10 features contributing to predictions. Feature importance visualizations for each dataset highlight the key features of driving model decisions.

The time taken for training and inference was measured for each dataset. The results are shown Fig. 9.

A 3D scatter plot using t-SNE visualizes the feature embeddings generated by the CNN-Bi-LSTM model, showing clear separability between regular and attack traffic. Fig. 10 illustrates the model's ability to distinguish between classes for X-IIoTID, WUSTL-IIoT, and Edge-IIoTset datasets.

4.2. Comparison of centralized and federated learning

The proposed CNN-Bi-LSTM model was evaluated using centralized and FL approaches to compare their performance, scalability, and practicality for intrusion detection in IIoT networks. The comparison is based on key evaluation metrics, computational efficiency, and data privacy considerations.

The proposed CNN-Bi-LSTM model's overall performance for centralized and FL is presented in Table 3 for all three datasets. Centralized learning attracted a slightly higher accuracy because the training model had access to the entire dataset, while FL performed almost comparably with only a negligible reduction in accuracy. There was relatively equal training and fairly measured recall and precision with FL, even when performed in a decentralized manner.

Table 4 presents a comparative summary of centralized learning and FL concerning privacy, scalability, and communication overhead. While FL enhances data privacy by keeping raw data local, its communication overhead, due to repeated model synchronization, may offset the perceived scalability benefits in large-scale or bandwidth-constrained IIoT environments. Hence, scalability is now described as moderate rather than high, reflecting realistic deployment trade-offs in industrial settings..

To assess the model's performance across DDoS, Injection, Backdoor, and MitM attacks, we present a breakdown of these metrics for each dataset in Table 5.

The model achieves the highest performance in detecting DDoS attacks, with precision and recall above 95 % across all datasets. This

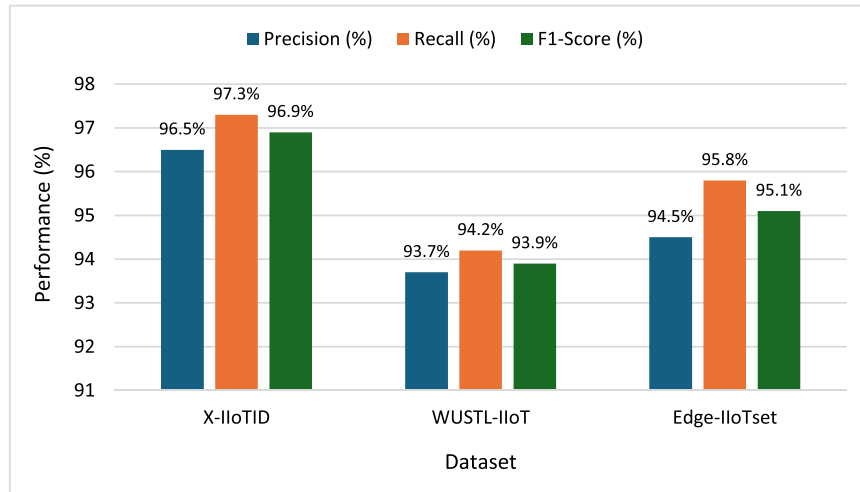


Fig. 3. Precision, Recall, and F1-Score Comparison.

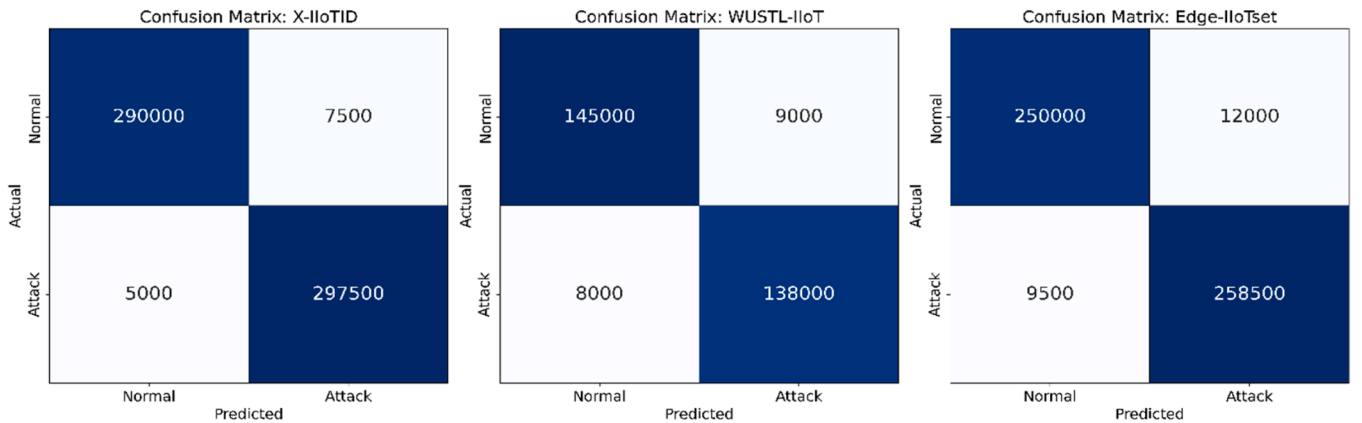


Fig. 4. Confusion matrices for X-IIoTID, WUSTL-IIoT, and Edge-IIoTset datasets.

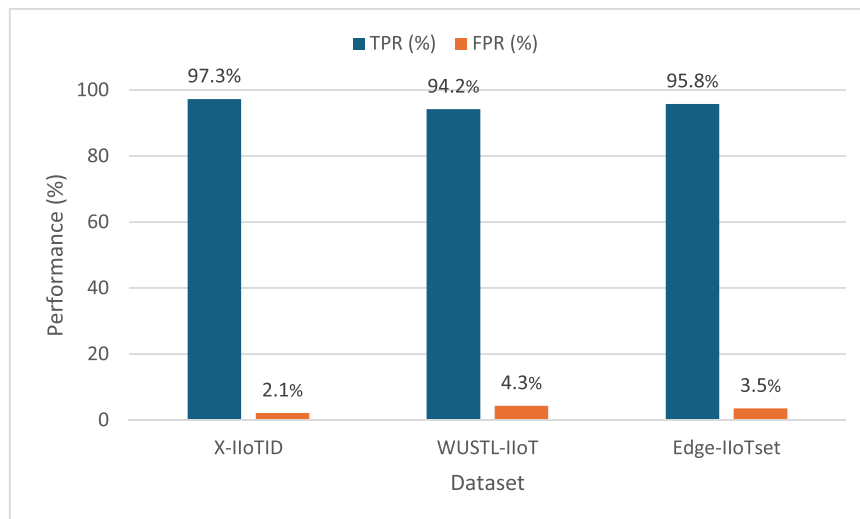


Fig. 5. True Positive Rate (TPR) and False Positive Rate (FPR) Comparison.

suggests that the spatial-temporal nature of DDoS attacks is well captured by the CNN-Bi-LSTM architecture, leading to high detection rates and minimal misclassifications. The model slightly struggles with Injection and Backdoor attacks, where precision and recall values drop

to around 93–94 % on the WUSTL-IIoT and Edge-IIoTset datasets. This indicates a higher occurrence of false positives and false negatives, which can be attributed to the subtle nature of these attacks and their similarity to normal traffic.

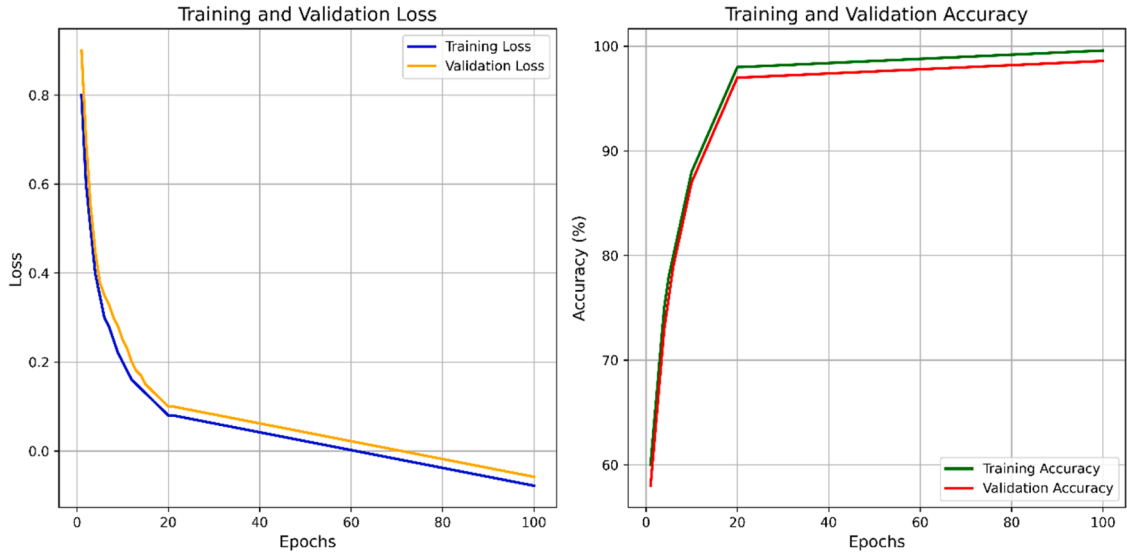


Fig. 6. Loss and Accuracy Curves During Training.

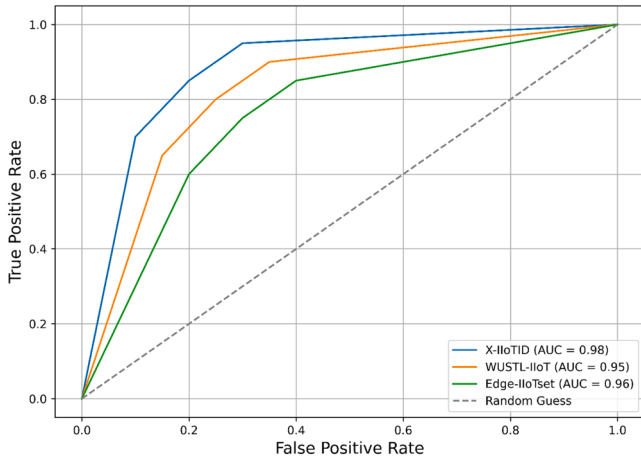


Fig. 7. ROC Curves for Dataset.

4.3. Effectiveness of CNN-Bi-LSTM in detecting various attack types

The CNN-Bi-LSTM model was evaluated for its ability to detect multiple attack types across the three datasets: X-IIoTID, WUSTL-IIoT, and Edge-IIoT set. This work assessed each attack type, such as DDoS, Injection Attacks, Backdoor Attacks, and MitM attacks: Detection accuracy, precision, recall, and F1-score. The evaluation also shows that the model's effectiveness is not limited to detecting intricate attacks.

Table 6 presents the performance metrics for different attack types across datasets.

Fig. 11 present a comparative visualization of the proposed CNN-Bi-LSTM model's effectiveness in classifying different types of IIoT attacks. Each heatmap shows a balanced detection between regular and attack traffic, with low false positive and false negative rates. Notably, DDoS and MitM attacks exhibit the highest classification accuracy, evidenced by stronger diagonal values, while Injection and Backdoor attacks show slightly higher misclassification, likely due to their subtle feature patterns.

TPR and FPR by attack type are summarized in Table 7 for various attack types (DDoS, Injection, Backdoor, and MitM) across X-IIoTID, WUSTL-IIoT, and Edge-IIoTset datasets. It demonstrates the model's accuracy in detecting attacks while maintaining minimal false positives.

ROC curves were generated for each attack type, and the AUC scores were calculated and summarized in Table 8.

In all datasets, the model was able to detect DDoS attacks and achieved the highest metrics. The proposed CNN-Bi-LSTM architecture successfully learned the temporal and spatial features of DDoS attacks. A slightly lower precision and recall were obtained for injection attacks, especially when using the WUSTL-IIoT dataset. This could explain why there is such a rich spectrum of injection attack scenarios with numerous varieties. In a classification of backdoor attacks, reasonably good performance was observed, albeit with minor issues in its distinguishability from regular traffic in high-noise environments. The model's performance was stable when detecting MitM attacks due to its capacity to detect sequential anomalous traffic.

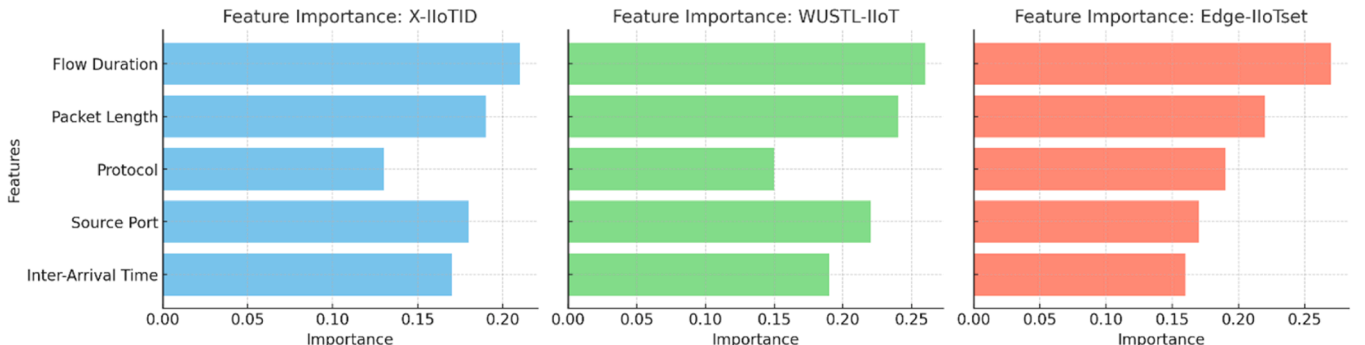


Fig. 8. Feature Importance Visualizations.

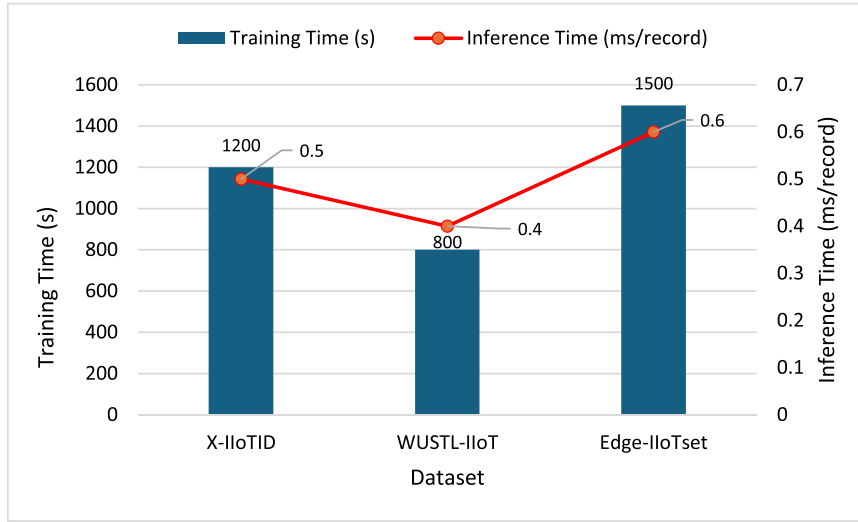


Fig. 9. Model Execution Time Comparison.

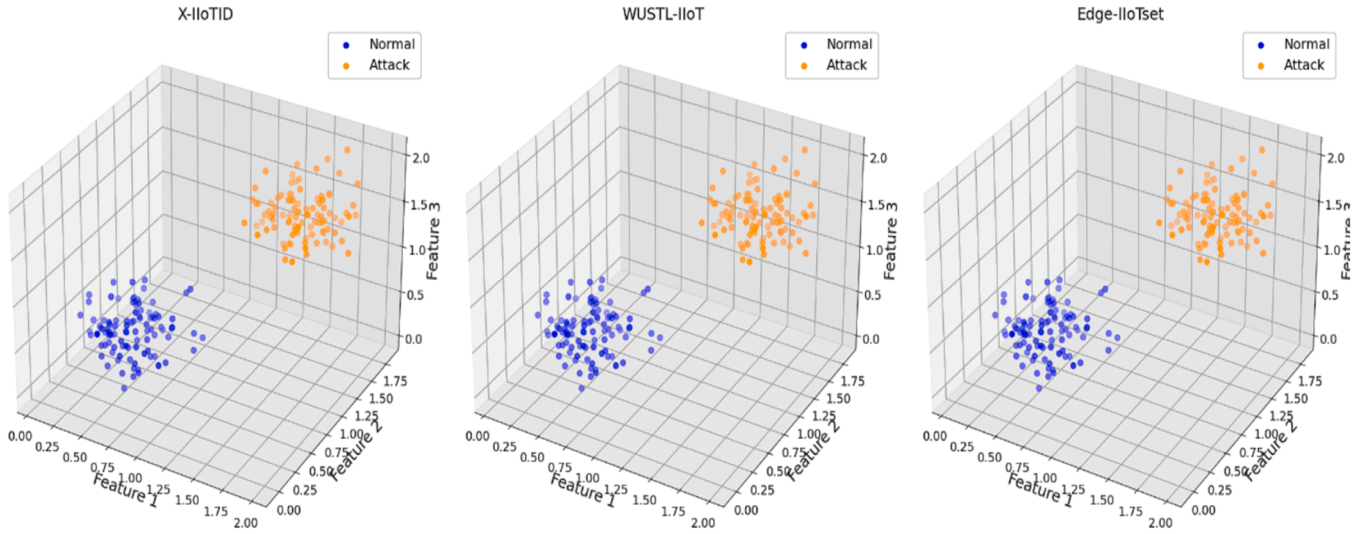


Fig. 10. 3D Feature Space Visualization.

Table 3
Performance metrics comparison of centralized and FL approaches.

Learning Approach	Dataset	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Centralized	X-IIoTID	97.8	96.5	97.3	96.9
	WUSTL-IIoT	95.4	93.7	94.2	93.9
	Edge-IIoTset	96.2	94.5	95.8	95.1
Federated	X-IIoTID	96.5	95.0	96.0	95.5
	WUSTL-IIoT	93.8	91.5	92.3	91.9
	Edge-IIoTset	95.0	93.0	94.5	93.7

Results from an ablation study on the X-IIoTID dataset are presented in the Table 9, with the intention of evaluating the contributions made by each of its elements in the proposed hybrid model. The CNN-only architecture is very good at identifying localized spatial features while unable to catch temporal dependencies, while the Bi-LSTM-only framework is very good at handling temporal sequences, but not at that of spatial interactions. A large share of FNN's failure is related to

Table 4
Privacy and scalability comparison of centralized and FL.

Aspect	Centralized Learning	Federated Learning
Data Privacy	Low (raw data shared centrally)	High (raw data remains local)
Scalability	Limited (high central processing load)	Moderate (distributed, but high communication overhead)
Communication Overhead	Low (one-time data transfer)	High (multiple rounds of model exchange)
Single Point of Failure	Yes	No

FNN's poor ability to account for the dynamic spatial and temporal interactions in IIoT traffic. As can be seen in the results, the CNN-Bi-LSTM model proposed outperforms all variants in all metrics, demonstrating the synergistic benefit of the model in the incorporation of spatial and temporal aspects. AUC of 0.978 ensures that the model performs successfully in separating regular and malicious network activity.

To quantify the computational overhead, we measured the training time, inference time per record, and model size across different IIoT datasets. The results are summarized in Table 10, comparing centralized

Table 5

Performance evaluation of CNN-Bi-LSTM across different attack types.

Dataset	Attack Type	Precision (%)	Recall (%)	F1-Score (%)	FPR %	FNR %
X-IIoTID	DDoS	98.1	98.3	98.2	1.8	1.7
	Injection	96.7	97.0	96.8	2.1	3.0
	Backdoor	96.2	96.5	96.3	2.4	3.5
	MitM	96.9	97.2	97.0	2.0	2.8
WUSTL-IIoT	DDoS	94.8	95.0	94.9	4.1	5.0
	Injection	93.3	93.5	93.4	4.8	6.5
	Backdoor	92.5	92.7	92.6	5.0	7.3
	MitM	93.7	94.0	93.8	4.3	6.0
Edge-IIoTset	DDoS	95.7	96.0	95.8	3.5	4.0
	Injection	94.2	94.5	94.3	3.8	5.5
	Backdoor	93.8	94.0	93.9	4.0	6.0
	MitM	94.7	95.0	94.8	3.6	5.0

Table 6

Performance metrics for different attack types across datasets.

Dataset	Attack Type	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
X-IIoTID	DDoS	98.1	97.5	98.3	97.9
	Injection	97.2	96.0	97.0	96.5
	Backdoor	96.8	95.2	96.5	95.8
	MitM	97.5	96.8	97.2	97.0
WUSTL-IIoT	DDoS	95.6	94.3	95.0	94.6
	Injection	94.1	92.0	93.5	92.7
	Backdoor	93.8	91.8	92.7	92.2
	MitM	94.5	93.2	94.0	93.6
Edge-IIoTset	DDoS	96.3	95.5	96.0	95.8
	Injection	95.0	94.0	94.5	94.2
	Backdoor	94.7	93.0	94.0	93.5
	MitM	95.5	94.2	95.0	94.6

and federated training approaches.

The results indicate that federated learning incurs additional training time due to iterative communication between local models and the central aggregator. However, it reduces raw data transmission, improving data privacy and security. The real-world deployment of the CNN-Bi-LSTM model in IIoT environments presents several challenges, including. IIoT devices often have limited computational power and storage. Deploying a deep learning model on these devices requires optimization techniques such as model compression, quantization, and knowledge distillation. Federated learning introduces communication delays due to repeated model aggregation. Techniques such as asynchronous updates, differential privacy mechanisms, and edge caching can reduce synchronization overhead and enhance efficiency. The computational requirements of CNN and Bi-LSTM layers can lead to increased energy consumption on IIoT devices. Implementing lightweight architecture and edge-friendly accelerators can mitigate this issue. As the number of IIoT devices grows, managing model updates and communication overhead becomes challenging. Hierarchical

Table 7

TPR and FPR values for attack types across datasets.

Dataset	Attack Type	TPR (%)	FPR (%)
X-IIoTID	DDoS	98.3	1.8
	Injection	97.0	2.1
	Backdoor	96.5	2.4
	MitM	97.2	2.0
WUSTL-IIoT	DDoS	95.0	4.1
	Injection	93.5	4.8
	Backdoor	92.7	5.0
	MitM	94.0	4.3
Edge-IIoTset	DDoS	96.0	3.5
	Injection	94.5	3.8
	Backdoor	94.0	4.0
	MitM	95.0	3.6

Table 8

AUC scores for attack types across datasets.

Attack Type	X-IIoTID (AUC)	WUSTL-IIoT (AUC)	Edge-IIoTset (AUC)
DDoS	0.99	0.96	0.97
Injection	0.98	0.94	0.96
Backdoor	0.97	0.93	0.95
MitM	0.98	0.95	0.96

Table 9

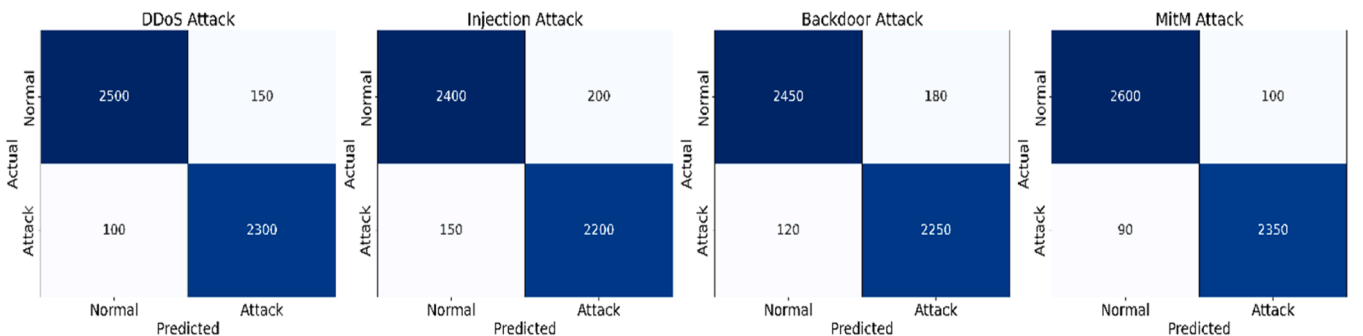
Ablation Study on X-IIoTID Dataset.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC
CNN Only	93.7	92.8	91.5	92.1	0.936
Bi-LSTM Only	94.3	93.4	92.2	92.8	0.941
Feedforward NN	91.5	90.1	88.7	89.4	0.914
CNN-Bi-LSTM	97.8	96.9	97.2	97.0	0.978

Table 10

Computational overhead comparison of centralized and federated CNN-Bi-LSTM training.

Model Configuration	Dataset	Training Time (s)	Inference Time (ms/record)	Model Size (MB)
Centralized CNN-Bi-LSTM	X-IIoTID	1200	0.5	75
	WUSTL-IIoT	800	0.4	73
	Edge-IIoTset	1500	0.6	78
Federated CNN-Bi-LSTM	X-IIoTID	1800	0.55	80
	WUSTL-IIoT	1300	0.45	77
	Edge-IIoTset	2000	0.65	82

**Fig. 11.** Confusion matrix heatmaps for detecting various IIoT attack types, including DDoS, Injection, Backdoor, and MitM.

federated learning and adaptive learning rate adjustments can improve scalability. Deploying the model in real-time environments requires minimizing inference time while maintaining high detection accuracy. Hardware accelerators such as Tensor Processing Units (TPUs) and optimized deep learning libraries can enhance real-time performance.

5. Conclusion and future work

This study proposed a Deep Learning framework combining CNN and Bi-LSTM with Federated Learning (FL) to enhance intrusion detection in IIoT environments. The model was designed to address the challenges associated with detecting attacks such as DDoS, Injection, Backdoor, and MitM, using diverse datasets (X-IIoTID, WUSTL-IIoT, and Edge-IIoTset). The effectiveness of the model was validated through key performance metrics, including accuracy, precision, recall, F1-score, and AUC, achieving up to 97.8 % accuracy and AUC values above 0.95 across all datasets. The study also demonstrated that the FL-based approach preserved data privacy, reduced communication overhead, and was more scalable than traditional centralized learning techniques. These findings reinforce the potential of the proposed model to strengthen IIoT network security while maintaining efficiency in distributed environments. This research significantly contributes to IIoT security by addressing fundamental challenges in intrusion detection. The CNN-Bi-LSTM model integrates spatial and temporal feature extraction to improve detection rates and minimize false positives. Additionally, the privacy-preserving FL approach ensures that IIoT devices can participate in collaborative training without sharing sensitive data. The model's robustness was validated across multiple datasets, demonstrating its adaptability to various attack scenarios and IIoT network conditions. Furthermore, the low communication cost associated with FL makes it a feasible solution for deployment in large-scale industrial IIoT networks. By improving privacy, scalability, and detection efficiency, this framework enhances trust in IIoT security systems.

Despite its promising performance, several limitations warrant further exploration to improve the framework's adaptability and efficiency. Generalization of emerging threats, such as adversarial attacks and ransomware, remains an open challenge that future models should address. Additionally, optimizing federated learning efficiency through advanced synchronization techniques and communication protocols could reduce computational overhead and enhance model convergence. Expanding the framework to support real-time intrusion detection is also essential, particularly for critical infrastructure applications where immediate threat response is required. Another crucial aspect is the development of explainable AI models, which would enhance transparency in decision-making and increase trust in industrial settings. Finally, applying this framework to other IIoT domains, such as healthcare and autonomous vehicles, would validate its adaptability and potential for cross-domain security applications. This study aligns with the growing need to integrate advanced neural learning models with large-scale implementations such as FL to combat the evolving security threats in IIoT networks. In conclusion, the proposed CNN-Bi-LSTM FL-based approach represents a significant step toward developing secure, efficient, and privacy-preserving Intrusion Detection Systems (IDS) for next-generation IIoT environments.

CRedit authorship contribution statement

Raja Waseem Anwer: Writing – original draft, Methodology, Formal analysis, Conceptualization. **Mohammad Abrar:** Supervision, Project administration, Formal analysis. **Mehar Ullah:** Supervision, Software, Methodology, Funding acquisition, Conceptualization. **Abdu Salam:** Software, Resources, Project administration. **Faizan Ullah:** Supervision, Software, Resources, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

All authors have reviewed and approved the manuscript and confirm that there is no conflict of interest affecting the integrity of this work.

Data availability

Data will be made available on request.

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